Analyzing the Impact of Advertising Spend on Sales

CMSE 202 Project Update – Machine Learning with the Advertising Dataset

Table of contents

01

Exploratory Data Analysis

See section details

02

Project Overview

See section details

03

Results & Evaluation

See section details

04

Model Training

See section details

05

Model Comparison

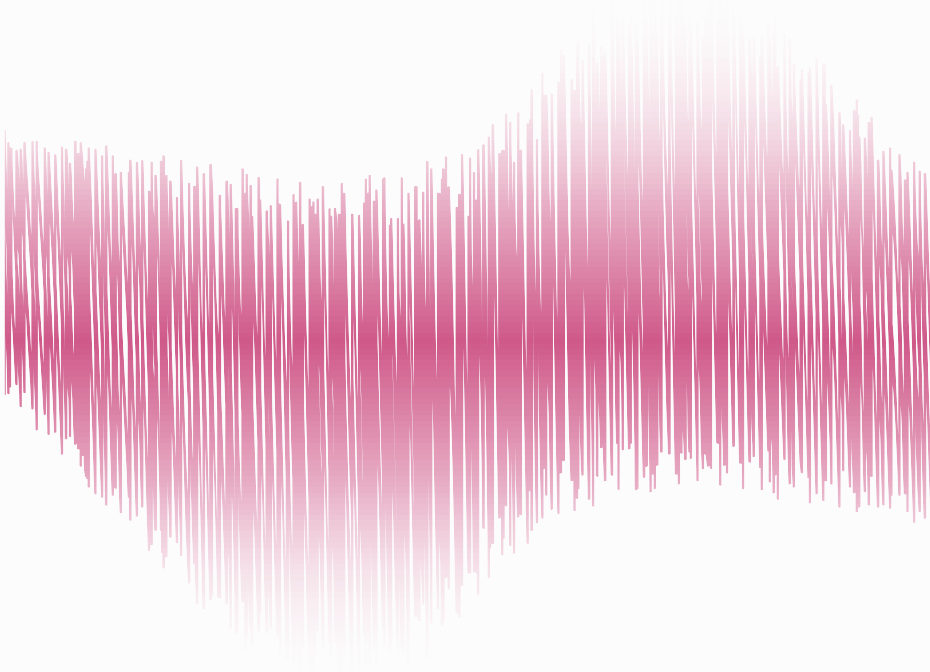
See section details

06

Conclusions

See section details

01
Project Overview



Guiding principles

Mercury

Mercury is the closest planet to the Sun

Saturn

Saturn is composed of hydrogen and helium

Jupiter

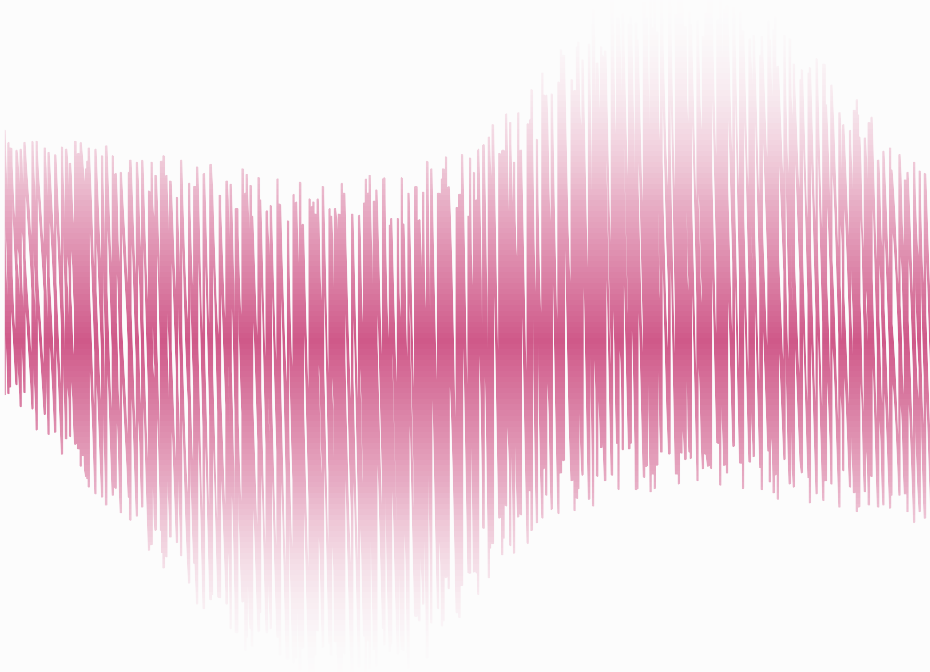
It's the biggest planet in the Solar System

Mars

Despite being red, Mars is a cold place

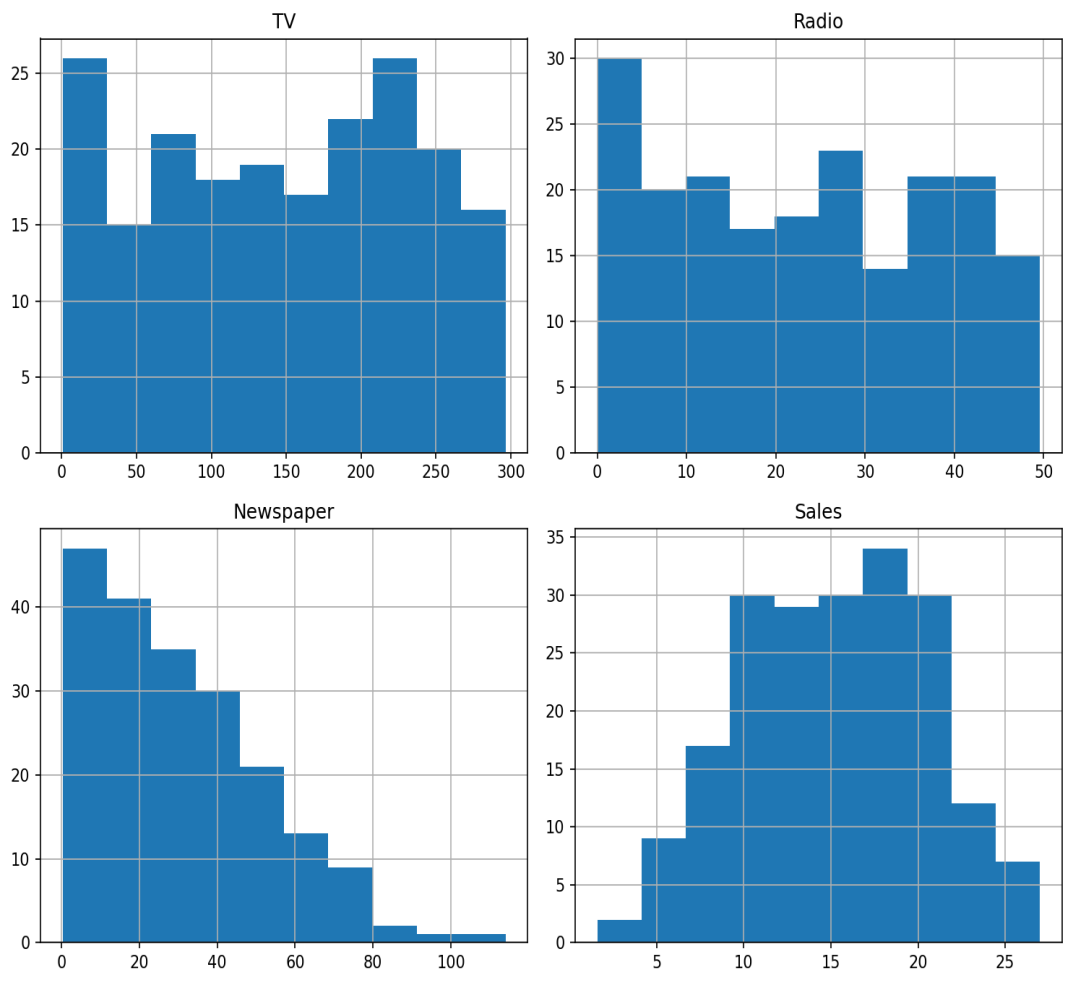
02 Exploratory Data Analysis

Distributions, scatter plots & correlations



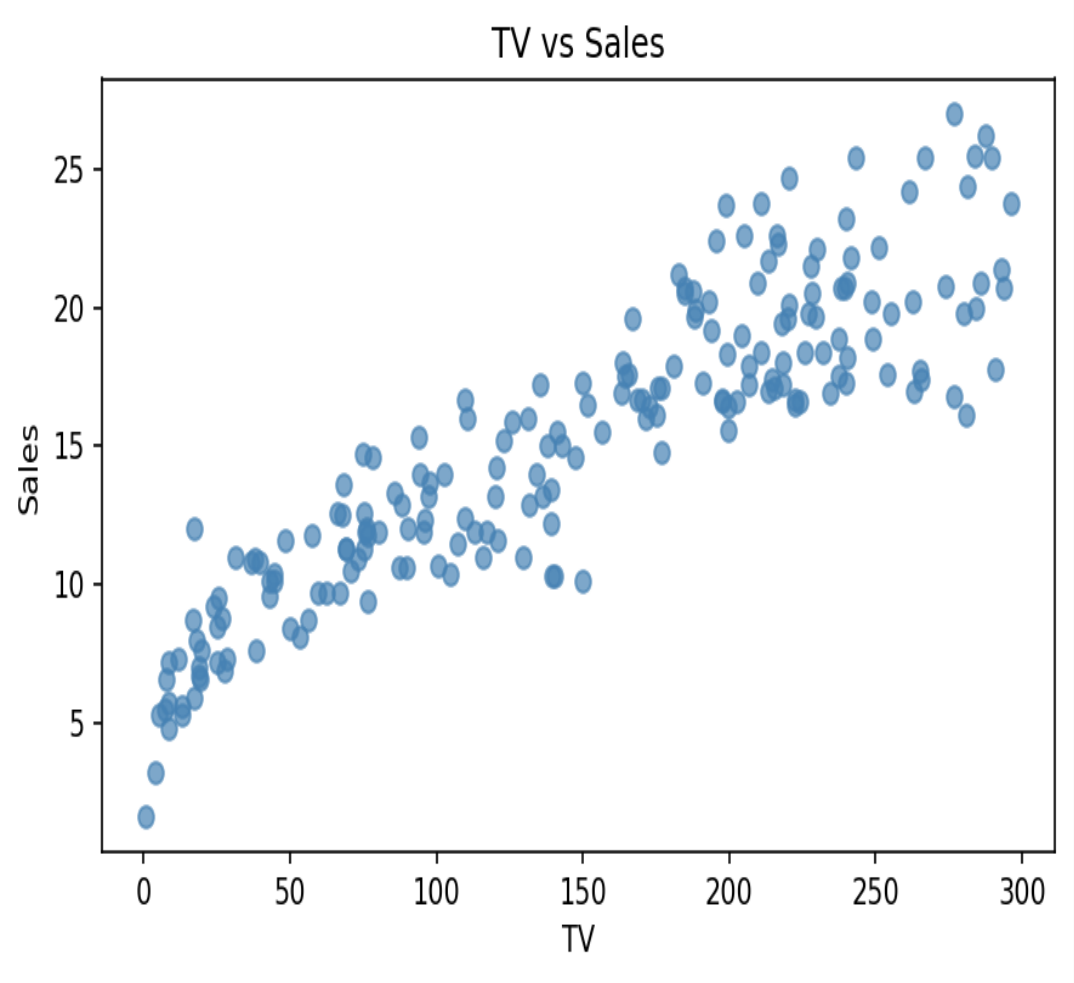
Feature Distributions

Histograms of all four variables: TV spend, Radio spend, Newspaper spend, and Sales. TV spending skews right, while Sales shows a roughly normal distribution.



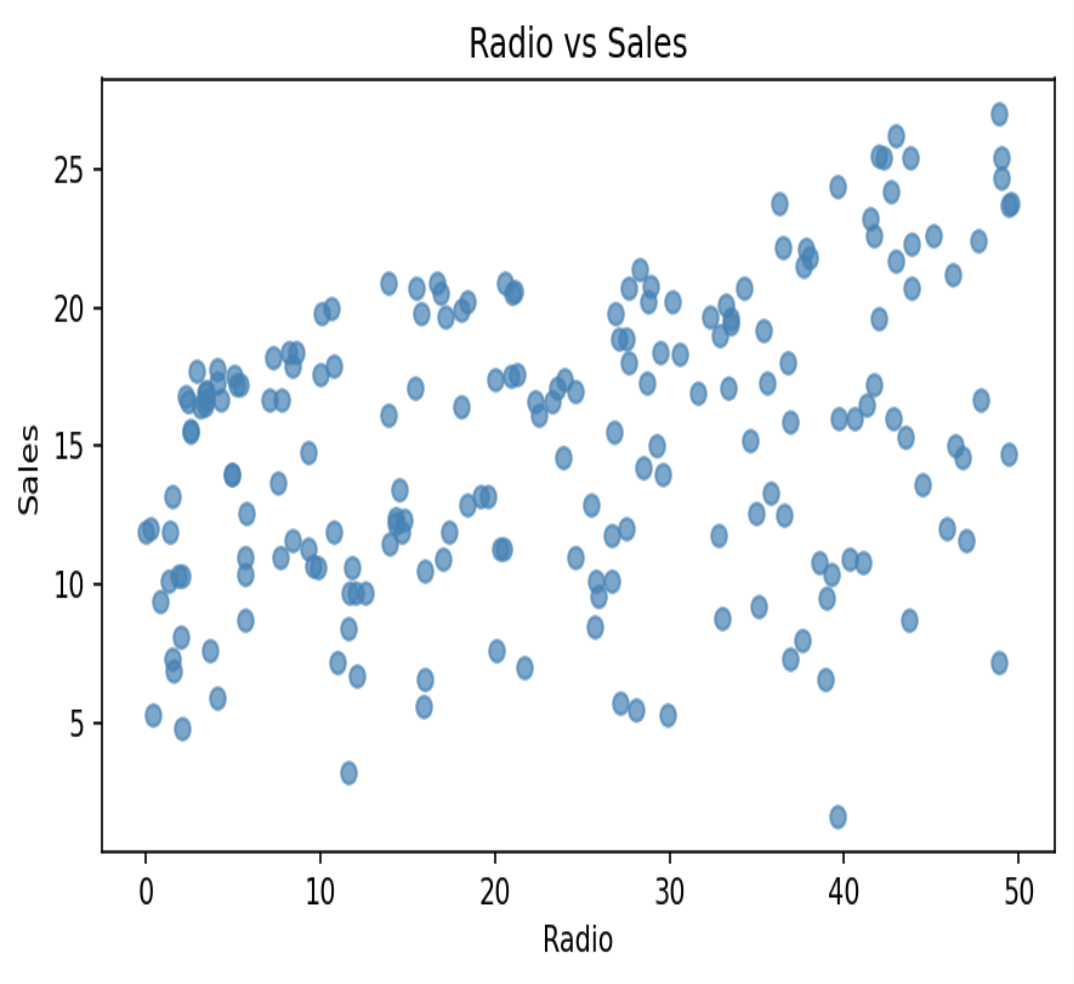
TV vs Sales

Strong positive linear relationship. TV advertising spend is the most correlated channel with sales, showing a clear upward trend across the range of spend values.



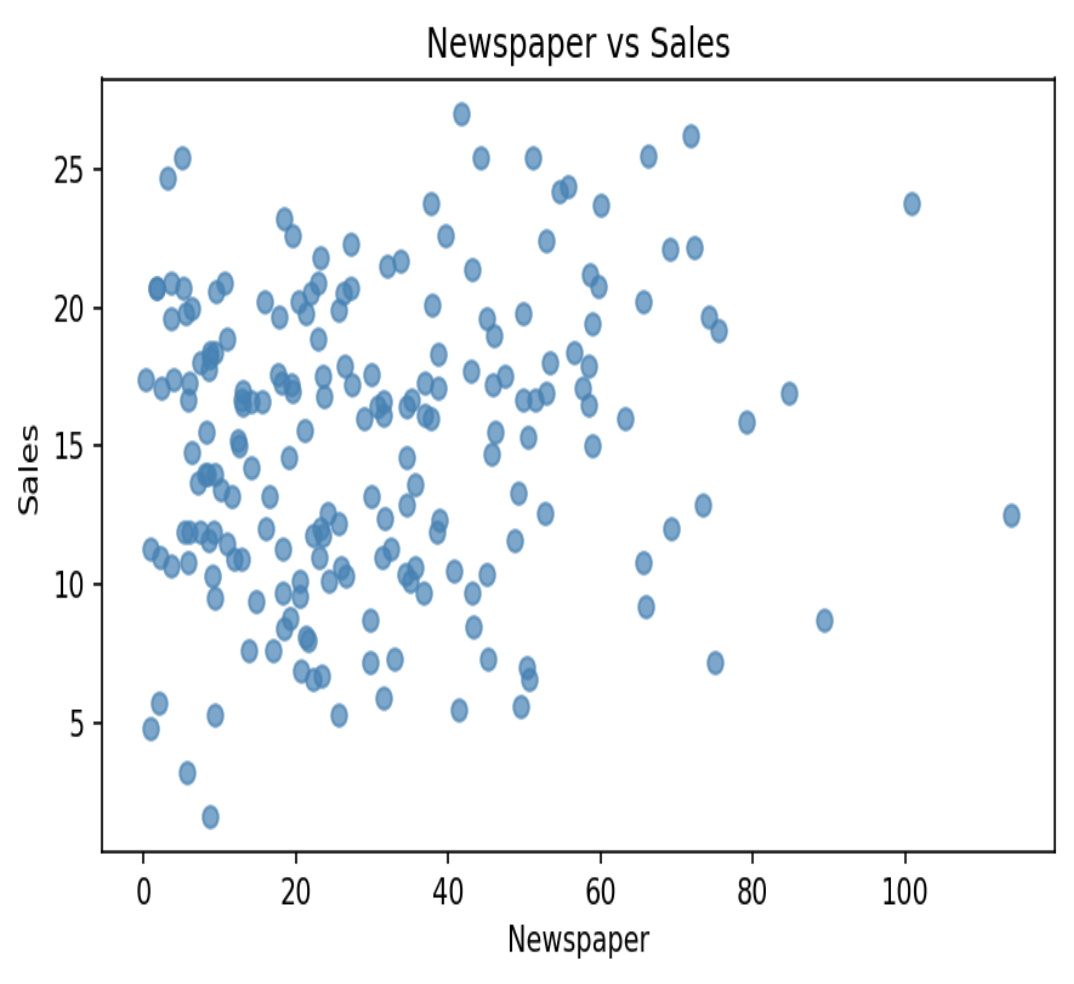
Radio vs Sales

Moderate positive relationship. Radio spend shows a meaningful but weaker correlation with sales compared to TV. Some spread in the data suggests other factors also influence sales.



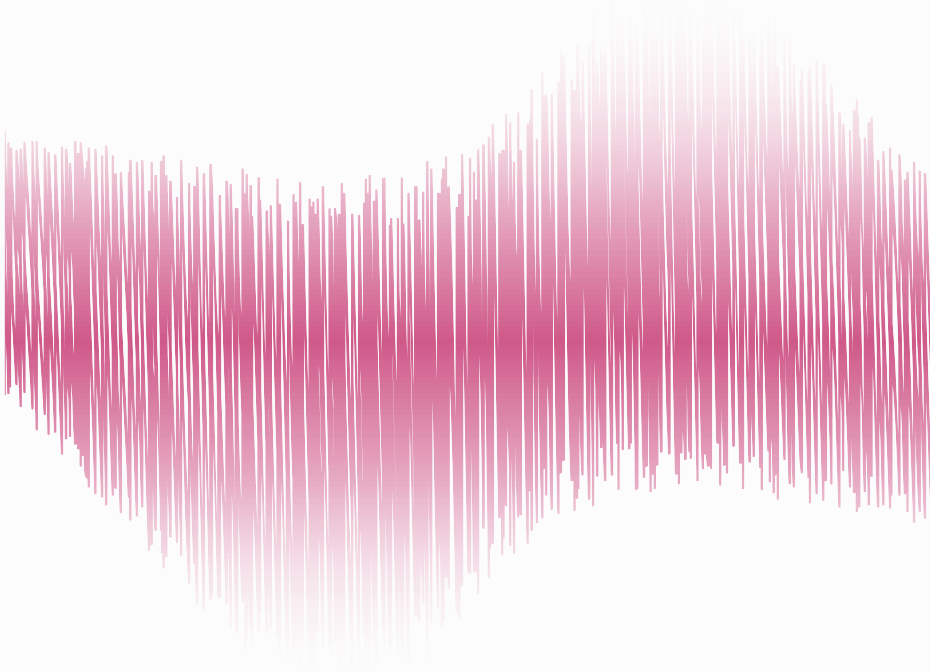
Newspaper vs Sales

Weak relationship. Newspaper spend shows the weakest correlation with sales among the three channels, with considerable scatter around any trend line.



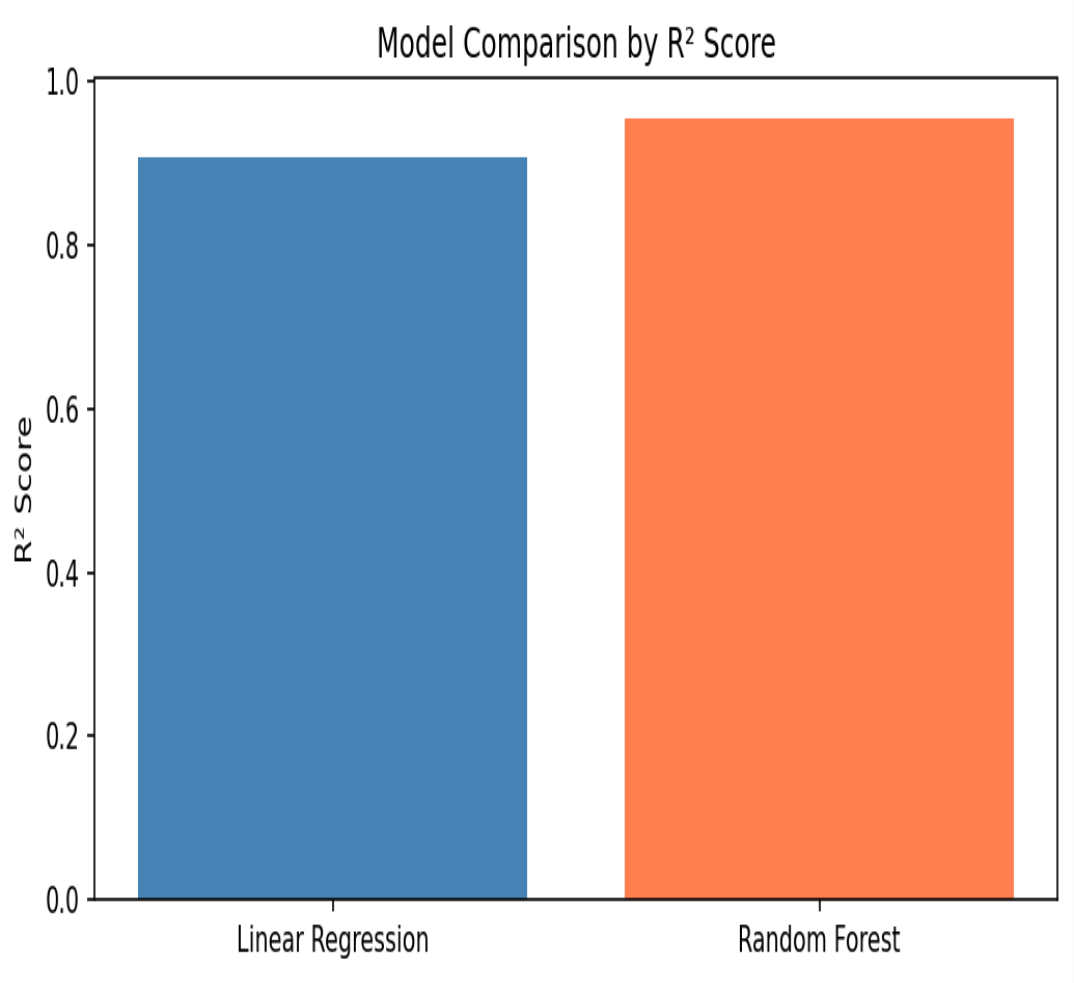
03 Model Training

Linear Regression vs.
Random Forest
Regressor



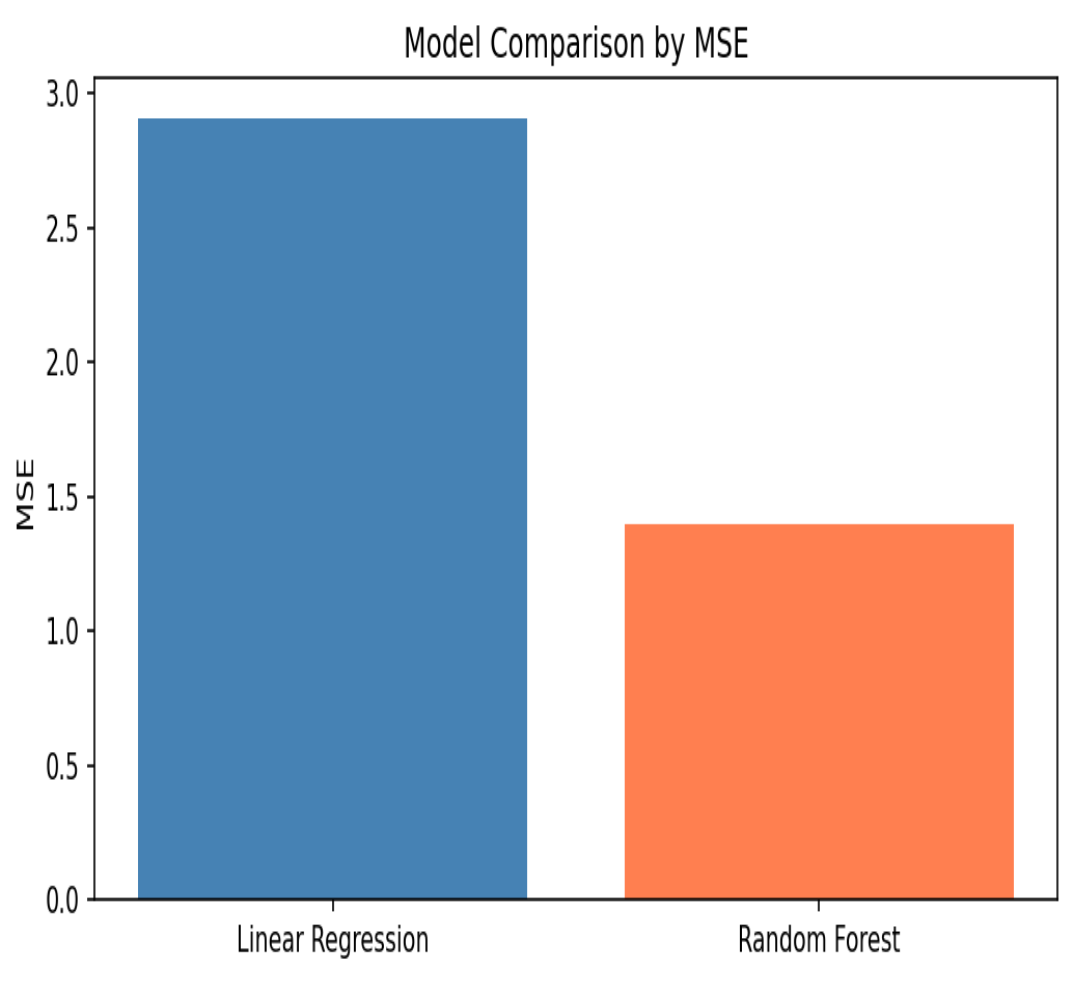
Model Comparison: R^2 Score

Random Forest ($R^2 = 0.955$) outperforms Linear Regression ($R^2 = 0.906$). Both models perform well, but the Random Forest captures non-linear relationships more effectively.



Model Comparison: MSE

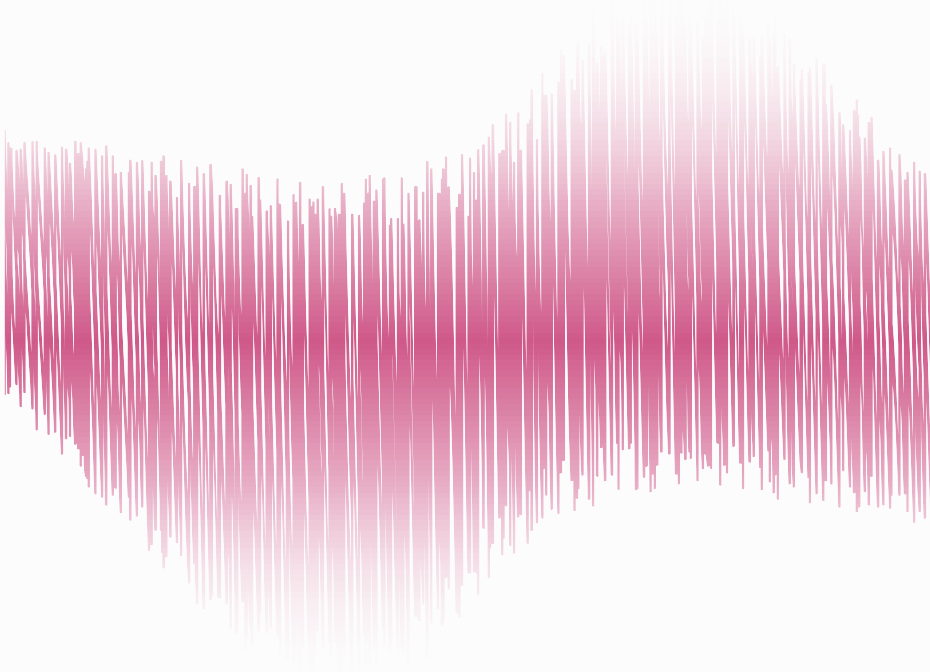
Random Forest achieves a lower Mean Squared Error (1.40) compared to Linear Regression (2.91), confirming it as the better-performing model for this dataset.



04

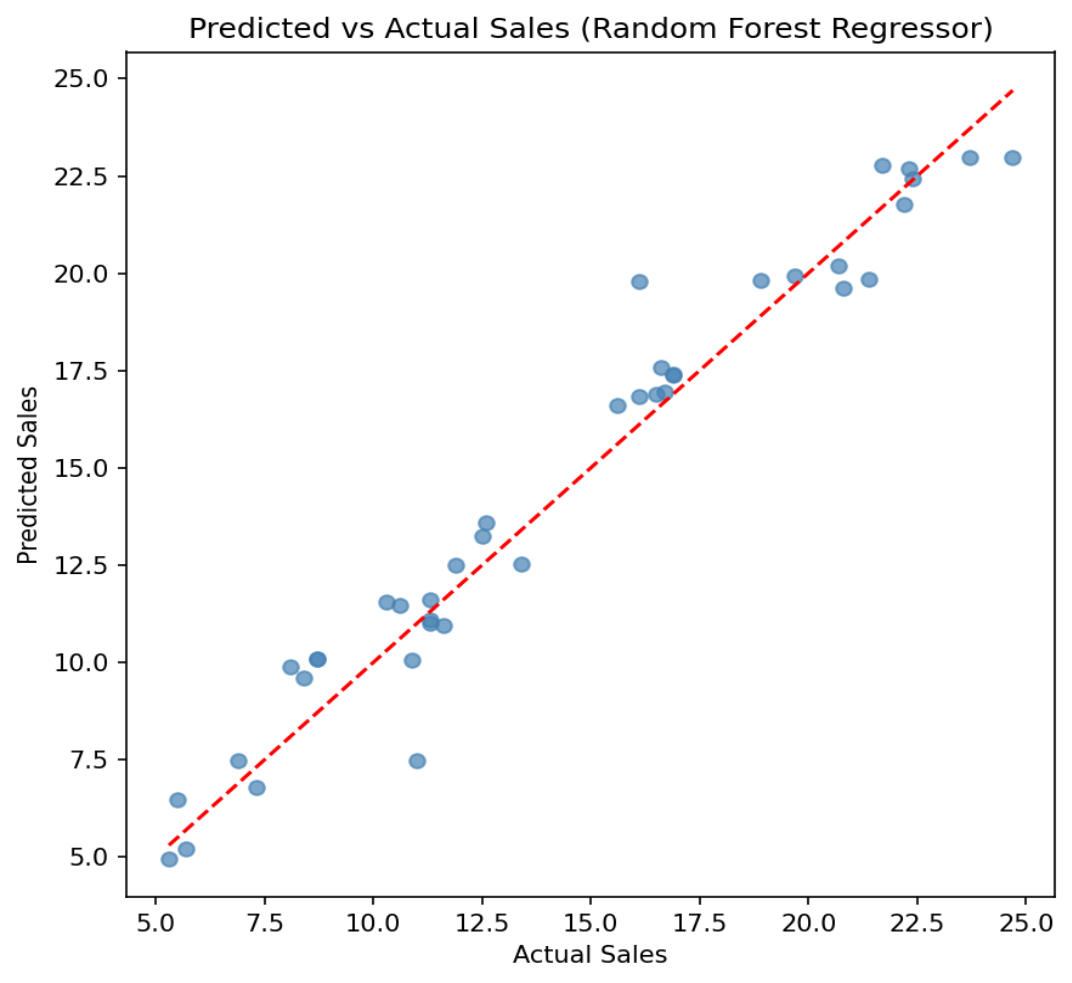
Results & Evaluation

Predictions, residuals & metrics performance



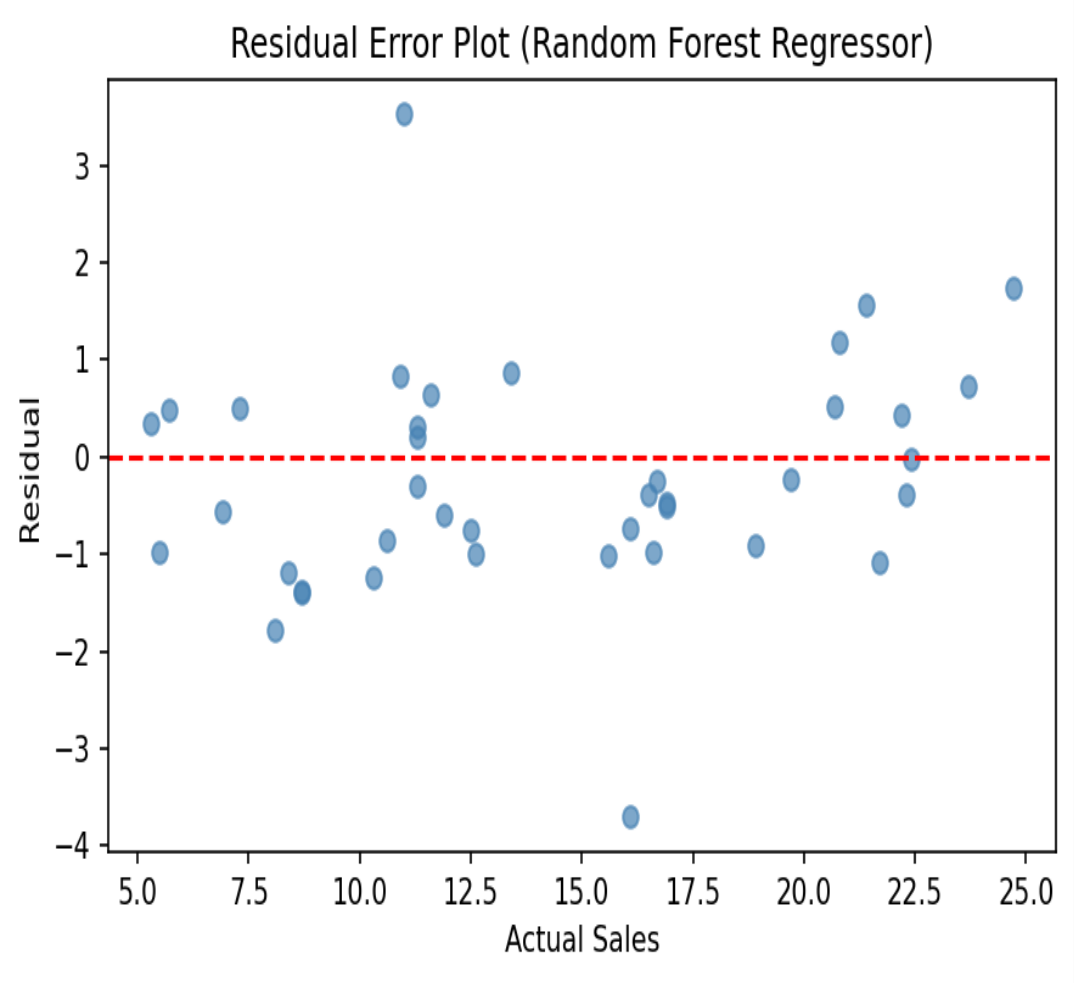
Predicted vs Actual Sales

Random Forest predictions closely follow the diagonal reference line, indicating strong model accuracy. Points near the line show the model generalizes well to unseen data.



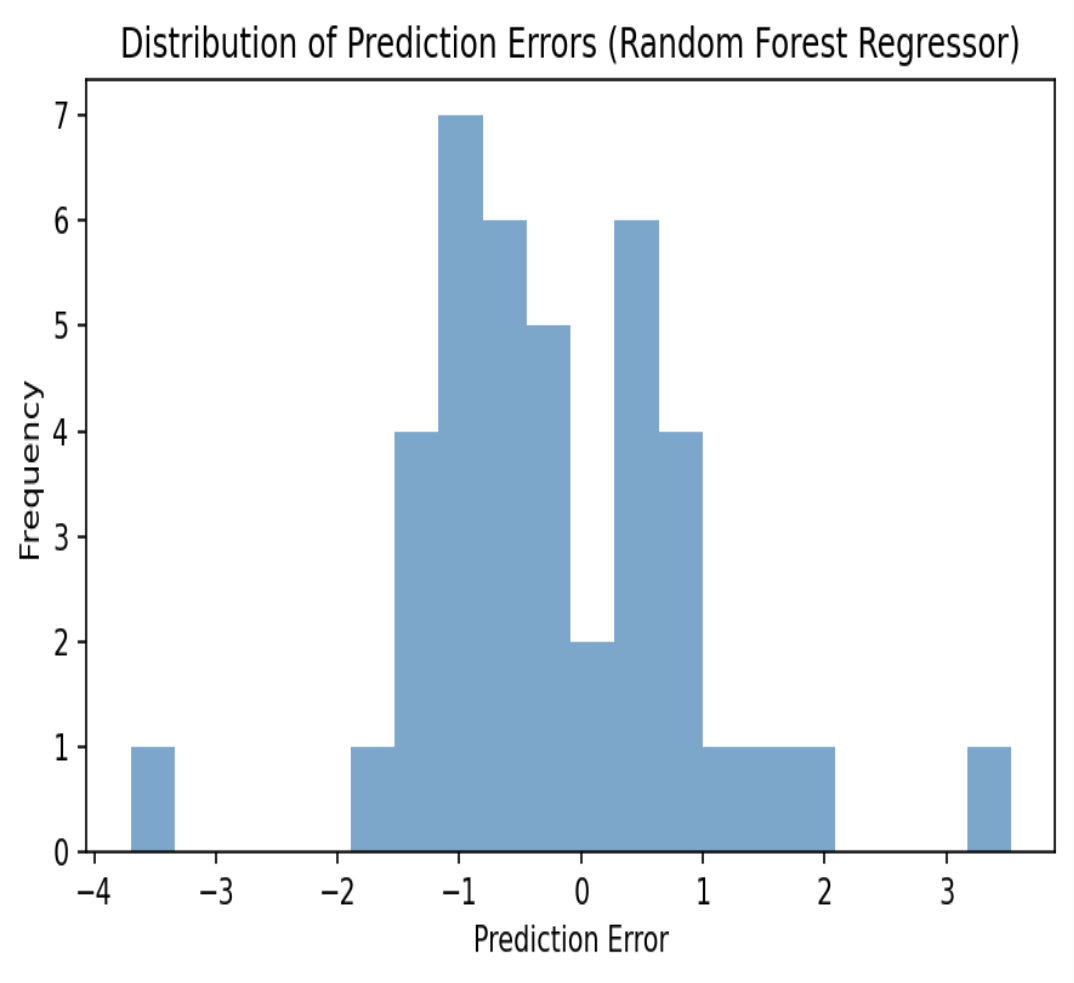
Residual Error Plot

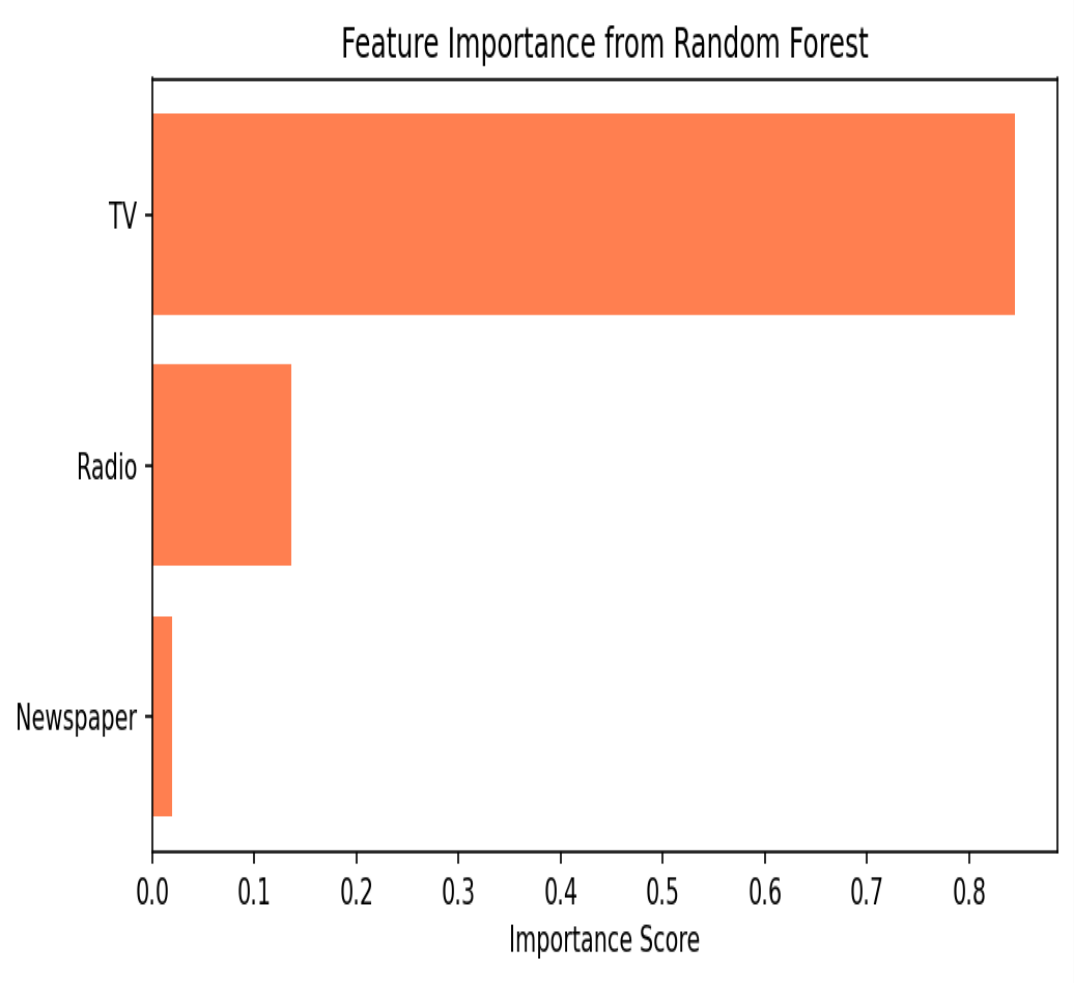
Residuals are randomly scattered around zero, suggesting the model does not have systematic bias. No clear pattern indicates the model fits the data well across all sales ranges.



Distribution of Prediction Errors

Prediction errors are roughly centered at zero and bell-shaped, confirming consistent model performance. A few large errors exist at the tails but most predictions are close to actual values.



Feature Importance (Random Forest)

TV advertising has by far the highest importance score (0.844), followed by Radio (0.136) and Newspaper (0.019). TV is the dominant predictor of sales in this dataset.